

Automated Trading System — NSE India
Built on Dhan API · Options · Futures · Equity

Broker	Dhan (dhanhq.co)
Exchange	NSE — Options, Futures, Equity
Strategies	Option Chain + SMC + Indicators + Price Action
Mode	Paper Trading (Switch to Live when ready)
Capital	■1,20,000 (configurable)
Alerts	Telegram Bot — @tradingrockstar_bot
Dashboard	http://127.0.0.1:8080
Version	1.0 — 05 Apr 2026

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1. System Architecture

The system follows a layered pipeline architecture. Each layer has a single responsibility and passes its output to the next layer.

Layer	Module	Responsibility
Data Layer	dhan_client.py	Fetch market data, option chain, place orders via Dhan API
Indicators	indicators.py	Compute EMA, RSI, MACD, Supertrend, ATR, VWAP, Volume
Option Chain	option_chain_analyzer.py	PCR, Max Pain, OI Walls, IV Skew, Call/Put Wall signals
Smart Money	smart_money.py	Order Blocks, FVG, BOS, CHoCH, Premium/Discount zones
Signal Engine	signal_aggregator.py	Score all sources (0-100), fire trade when confidence $\geq 60\%$
Risk Manager	risk_manager.py	Position sizing, daily loss limit, trailing stops, P&L log
Order Executor	order_executor.py	Place entry+SL orders, monitor positions, EOD square-off
Orchestrator	main.py	Run full loop every 60s during market hours (9:20–3:15)
Dashboard	dashboard/app.py	Live web UI on http://127.0.0.1:8080

2. Module Overview

config/settings.py

Central configuration file. All risk rules, indicator periods, strategy toggles, API keys (via .env), Telegram settings, and market hours are defined here. This is the first file to edit when tuning the system.

data/dhan_client.py

Full Dhan REST API v2 wrapper. Covers: LTP, OHLC, historical candles (1/5/15/60 min), option chain with all Greeks, order placement/modification/cancellation, positions, holdings, fund limits.

data/indicators.py

Pure Python indicator engine (no TA-Lib required). Includes: EMA (9/21/50/200), SMA, WMA, RSI, MACD, Stochastic, CCI, ATR, Bollinger Bands, Keltner Channel, Supertrend, ADX, VWAP, OBV, VPT, Pivot Points, Swing Highs/Lows.

data/option_chain_analyzer.py

Parses Dhan option chain data and computes: PCR (OI & Volume), Max Pain, Call Wall / Put Wall (OI-based S/R), IV Skew, OI Buildup/Unwinding detection. Outputs: BUY_CE / BUY_PE / SELL_CE / SELL_PE / WAIT with 0-100 strength score.

strategies/smart_money.py

Smart Money Concepts engine. Detects: Bullish/Bearish Order Blocks, Fair Value Gaps (FVG), Break of Structure (BOS), Change of Character (CHoCH), Liquidity Sweeps, Premium/Discount zones. Generates BUY/SELL with precise SL & TP levels.

strategies/signal_aggregator.py

Combines all signal sources into one final decision. Scoring: Option Chain (35pts) + Indicators (35pts) + SMC (20pts) + Price Action (10pts). Trade fires only when score ≥ 60 AND R:R ≥ 1.5 .

risk/risk_manager.py

Enforces all risk rules: max 2% capital per trade, 3% daily loss limit, 5 max positions, trailing stops, position sizing (Fixed%/Kelly/Fixed Lots). Logs every trade to logs/trades.csv with full entry/exit details.

execution/order_executor.py

Places orders via Dhan API: entry (LIMIT/MARKET), SL (STOP_LOSS_MARKET), target (LIMIT). Monitors all open positions on every tick. Sends Telegram alerts. Auto squares off all intraday positions before 3:15 PM.

3. Strategies Explained

3.1 Option Chain Analysis

Analyses the full NSE option chain for NIFTY and BANKNIFTY every 60 seconds.

Metric	What It Tells Us	Signal
PCR > 1.2	More puts than calls = hedging = support	Bullish
PCR < 0.8	More calls = complacency or greed	Bearish
Call Wall	Highest call OI strike = resistance	Sell above
Put Wall	Highest put OI strike = support	Buy near
Max Pain	Where option writers profit most	Range target
OI Buildup	Rising OI + rising price = long buildup	Trend follow
IV Skew	Put IV > Call IV = smart money hedging	Caution/Hedge

3.2 Technical Indicators

Indicator	Period	Signal Condition
EMA Stack	9/21/50	9 > 21 > 50 = Bullish 9 < 21 < 50 = Bearish
RSI	14	< 30 = Oversold (Buy) > 70 = Overbought (Sell)
MACD	12/26/9	Histogram cross above 0 = Bullish below 0 = Bearish
Supertrend	10, 3.0	Price above line = Uptrend below = Downtrend
ADX	14	> 25 = Strong trend < 20 = Weak / Range
Bollinger	20, 2	Price at lower band = Oversold upper band = Overbought
VWAP	Intraday	Price above VWAP = Bullish below = Bearish
Volume	20 MA	Spike > 2x average = Confirms breakout/breakdown

3.3 Smart Money Concepts (SMC)

Order Block (OB): Last bearish candle before a bullish impulse (Bullish OB) or last bullish candle before a bearish impulse (Bearish OB). Price returning to OB = high probability entry.

Fair Value Gap (FVG): 3-candle imbalance where middle candle leaves a gap. Price tends to return to fill the gap. Minimum gap size: 0.1% of price.

Break of Structure (BOS): Continuation signal — price breaks the last swing high (bullish BOS) or last swing low (bearish BOS). Confirmed after 2 candles.

Change of Character (CHoCH): Reversal signal — price breaks against the current structure. Signals a potential trend change. Wait for confirmation before trading.

Premium/Discount: Premium zone = top 50% of swing range (sell area). Discount zone = bottom 50% (buy area). Best entries in discount for buys, premium for sells.

4. Risk Management Rules

Rule	Setting	What Happens When Triggered
Max Capital Per Trade	2%	Max ■2,400 at risk per trade (on ■1.2L capital)
Daily Loss Limit	3%	Trading stops for the day if P&L drops by ■3,600
Max Open Positions	5	No new trades until an existing one closes
Max Positions / Symbol	2	Prevents over-concentration in one stock
Max Trades Per Day	20	Prevents overtrading / revenge trading
Min Risk:Reward	1.5x	Trade rejected if target < 1.5x the risk
Trailing Stop Trigger	+1.5%	Trailing stop activates after 1.5% profit
Trailing Stop Distance	0.75%	Stop trails 0.75% below the highest price reached
EOD Square Off	3:15 PM	All F&O positions closed automatically before expiry

■ All risk settings can be changed in `config/settings.py` under the `RiskConfig` dataclass. It is strongly recommended to run in paper trading mode for at least 2 weeks before going live.

5. Configuration Guide

5.1 .env File (Credentials)

Located in the trading_system/ folder. Never share this file.

```
DHAN_CLIENT_ID=your_client_id
DHAN_ACCESS_TOKEN=your_access_token
PAPER_TRADING=true
TELEGRAM_BOT_TOKEN=your_bot_token
TELEGRAM_CHAT_ID=your_chat_id
```

5.2 Key Settings in config/settings.py

Setting	Default	Description
PAPER_TRADING	true	Set to false only when ready for live trading
max_capital_per_trade_pct	2.0%	% of total capital at risk per trade
max_daily_loss_pct	3.0%	Stop trading if daily loss exceeds this %
max_open_positions	5	Maximum simultaneous open trades
risk_reward_min	1.5	Minimum R:R before a trade is taken
trailing_stop_enabled	True	Enable/disable trailing stop
pcr_bullish_threshold	1.2	PCR above this = bullish signal
pcr_bearish_threshold	0.8	PCR below this = bearish signal
SIGNAL_THRESHOLD	60	Min confidence score (0-100) to fire a trade
AVOID_TRADE_NEAR_OPEN_MINS	5	Skip trading in first 5 min after open
AVOID_TRADE_NEAR_CLOSE_MINS	15	Stop new trades 15 min before market close

6. Running the System

Step 1 — Install Dependencies

```
/Library/Developer/CommandLineTools/usr/bin/python3 -m pip install -r requirements.txt
```

Step 2 — Test Connections

```
/Library/Developer/CommandLineTools/usr/bin/python3 check_connection.py
```

```
/Library/Developer/CommandLineTools/usr/bin/python3 test_telegram.py
```

Step 3 — Run Paper Trading (Default)

```
/Library/Developer/CommandLineTools/usr/bin/python3 main.py
```

Starts the full trading loop. Runs 9:20 AM – 3:15 PM IST on weekdays.

Step 4 — Open Dashboard

```
/Library/Developer/CommandLineTools/usr/bin/python3 dashboard/app.py
```

Open Chrome → <http://127.0.0.1:8080>

Step 5 — Go Live (When Ready)

```
/Library/Developer/CommandLineTools/usr/bin/python3 main.py --live
```

■ *This will place REAL orders on your Dhan account. Type CONFIRM when prompted. Ensure paper trading has been profitable for 2+ weeks first.*

Stop the Engine

Press Ctrl+C in the terminal. The engine will square off all open positions automatically.

7. Dashboard Guide

URL: <http://127.0.0.1:8080> (auto-refreshes every 30 seconds)

Panel	Shows
Today's P&L	Net profit/loss for the session in ■ and %
Open Positions	All live trades with entry, LTP, unrealised P&L, SL
Trades Today	Count of total, closed trades and daily trade limit
Win Rate	Winners vs Losers with percentage
Available Capital	Remaining deployable capital with visual bar
Daily Loss Meter	How much of the 3% daily loss limit has been used
System Status	Dhan API, Telegram, Risk Manager, active strategies
Recent Signals	Last 10 signals with confidence bars and R:R
Trade History	Last 20 closed trades with entry, exit, P&L, status

8. Telegram Alerts

Bot: @tradingrockstar_bot — You receive alerts for:

- ■ Trade Entry — Symbol, type, entry price, SL, target, confidence, R:R
- ■ Stop Loss Hit — Exit price, P&L, reason
- ■ Target Hit — Exit price, profit amount
- ■ Trailing Stop Update — New stop level
- ■ EOD Square Off — All positions closed with summary
- ■ Daily Summary — Total trades, win rate, net P&L;
- ■ Engine Errors — Any critical errors during runtime

9. Monday Morning Checklist

- Ensure internet connection is stable
- Open terminal in trading_system/ folder
- Run: `python3 main.py` (starts at 9:15 AM)
- Open Chrome → `http://127.0.0.1:8080` for dashboard
- Open Telegram → watch for @tradingrockstar_bot alerts
- Monitor first 30 minutes manually to verify signals
- Check logs/trading_engine.log if anything looks wrong
- Do NOT interrupt the process during market hours
- After market close — review logs/trades.csv for results

10. Fixing Dhan Security IDs

The instrument security IDs in main.py need to match Dhan's official instrument list for market data to work correctly. Follow these steps:

1. Log in to dhanhq.co → API section → Download Instruments CSV
2. Open the CSV and search for NIFTY, BANKNIFTY, RELIANCE, HDFCBANK, INFY etc.
3. Copy the SEM_SMST_SECURITY_ID value for each symbol
4. Open trading_system/main.py → find INSTRUMENT_REGISTRY
5. Update each "security_id" value with the correct ID from the CSV
6. Save the file and restart main.py

11. Troubleshooting

Error / Issue	Fix
<code>pip: command not found</code>	Use: <code>python3 -m pip install -r requirements.txt</code>
Port 5000 already in use	Dashboard uses port 8080 now. Or disable AirPlay in System Settings
API returns 401	Regenerate Access Token on dhanhq.co — tokens expire
API returns 400 Invalid Request	Update security IDs in <code>main.py</code> from Dhan instruments CSV
No data returned for symbol	Check <code>security_id</code> is correct and Data API subscription is active
Telegram message not received	Send <code>/start</code> to your bot first, then run <code>test_telegram.py</code> again
Engine not trading at 9:15	Trading starts at 9:20 (skips first 5 min). Check <code>logs/</code> folder
<code>ModuleNotFoundError</code>	Re-run: <code>python3 -m pip install -r requirements.txt</code>
Daily loss limit hit	Normal — trading stops for the day. Resumes next trading day
Signal shows WAIT always	Confidence threshold is 60%. Lower it in <code>signal_aggregator.py</code> if needed